



Voltage Divider & Potentiometer: Manual Dimmer Fun!

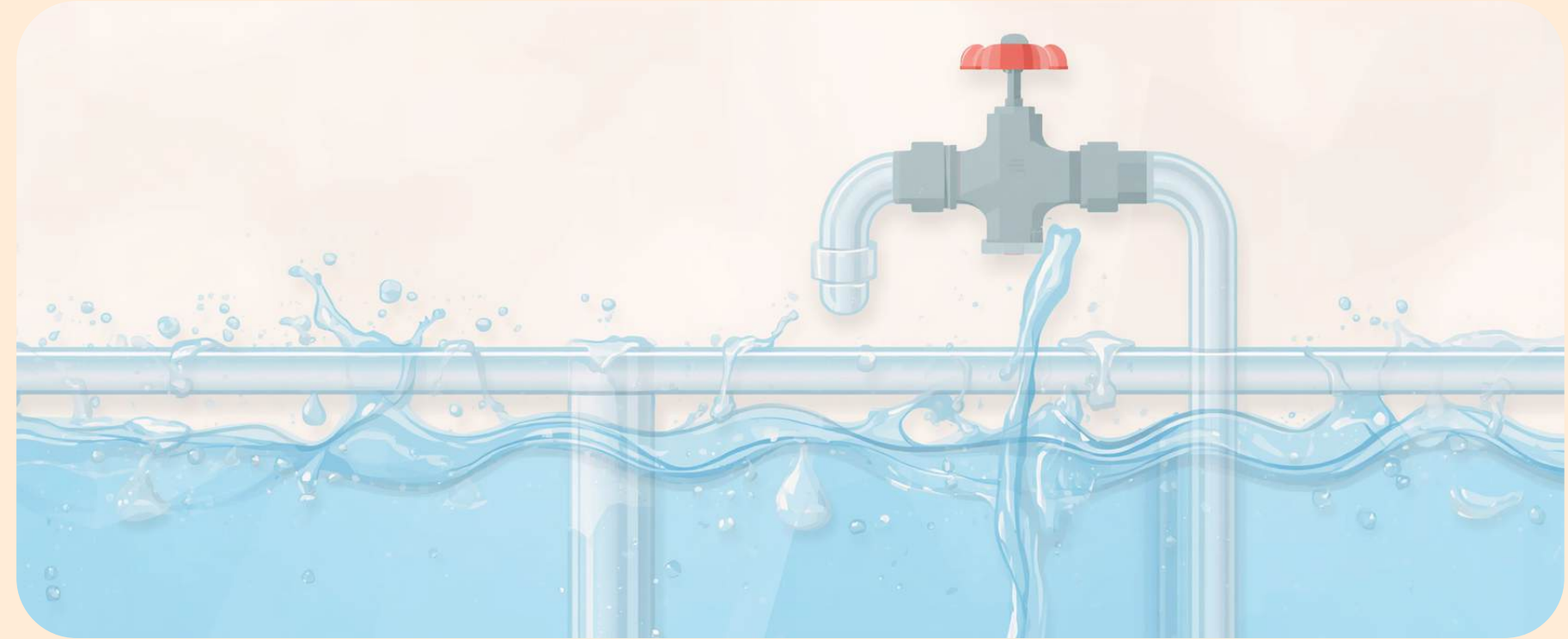




Voltage Divider & Potentiometer: Manual Dimmer Fun!

Imagine Electricity Flowing Like Water!

The Water Pipe Analogy



Just like turning this knob changes fan speed, we can control electricity! Turn the knob more for more flow, turn it less for less flow. Cool, right?

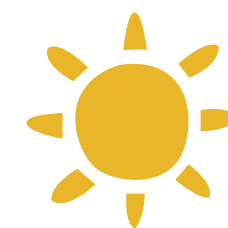


What is Electricity?

Powering Our World!



Electricity helps power our lights, gadgets, and so much more!



Electricity flows like water through pipes, moving from one place to another.

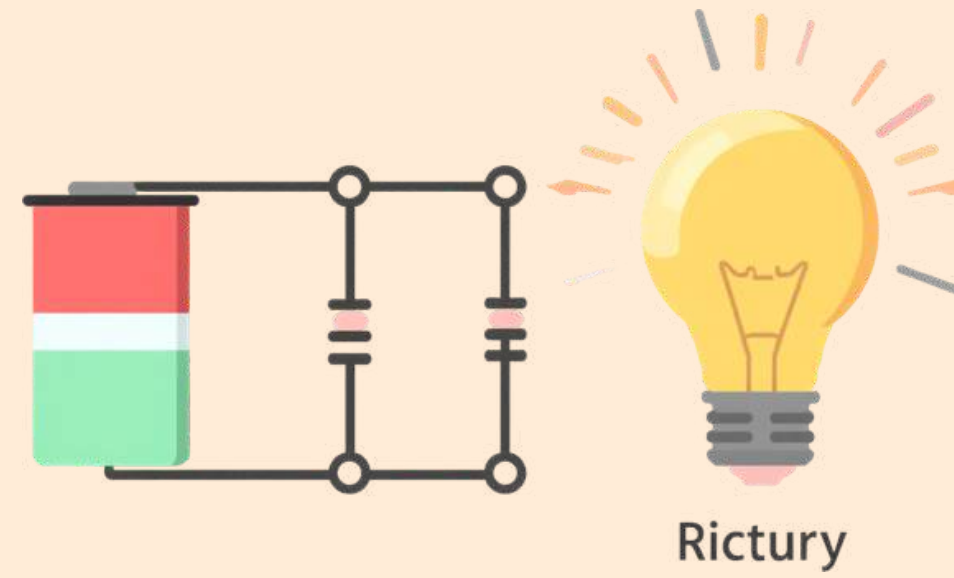


Electricity needs a path to flow, just like water needs a pipe!

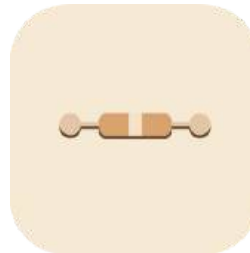


What is a Voltage Divider?

Splits Electricity to Change Brightness!



Battery: The power source that gives electricity energy!



Resistors in Series: Two resistors connected one after another!



Light Bulb: Gets brighter or dimmer based on the split!



What is a Potentiometer?

The Magic Knob!



It's a special knob you can turn to change how much electricity flows!



Turn the knob clockwise to let more electricity flow through!



Use it to make your light brighter or dimmer!





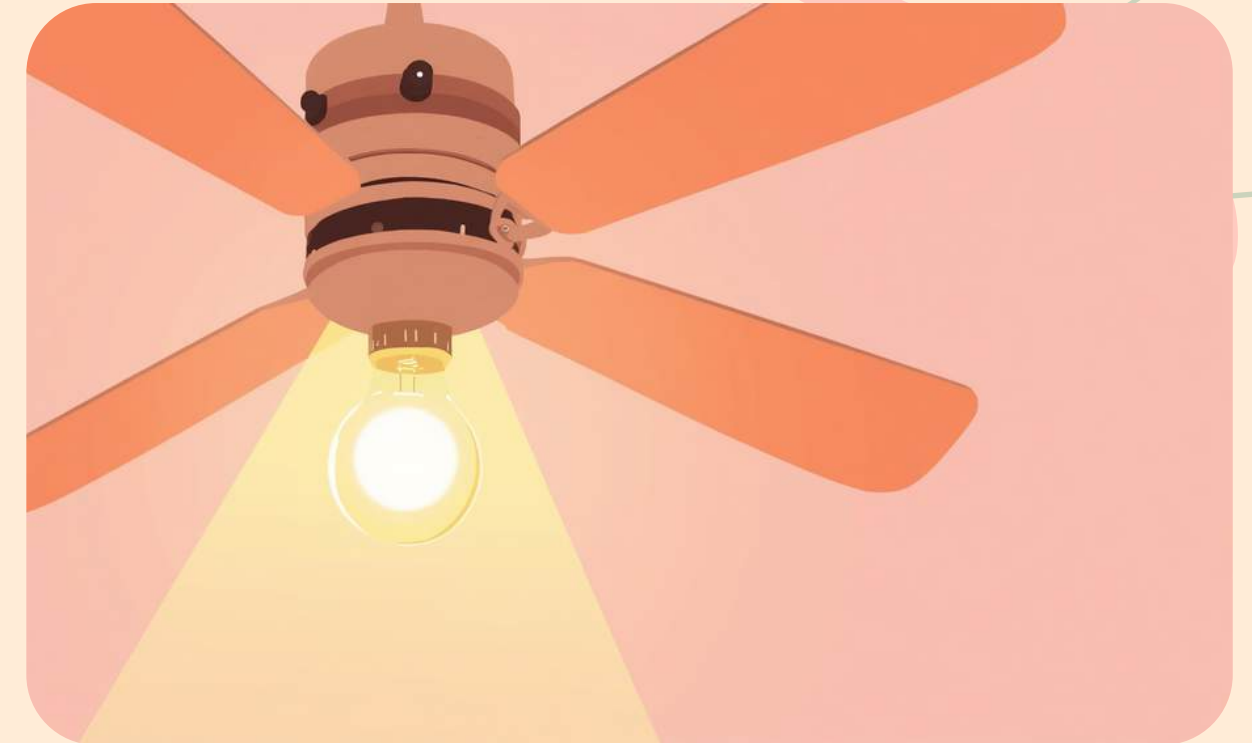
Voltage Divider & Potentiometer

How Does a Potentiometer Work as a Dimmer?

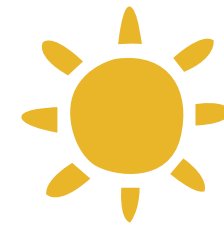


The Dimmer Effect

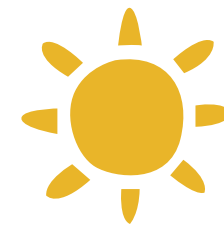
Turning the potentiometer knob changes the brightness of the light! More turn = more light! Less turn = less light!



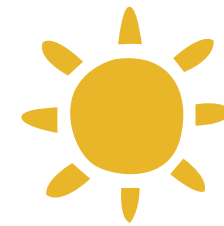
Hands-On Experiment!



Connect the battery, potentiometer, and light bulb exactly as shown!



Turn the knob slowly and watch the light get brighter or dimmer!



Ask your teacher or family member for help before starting!

Quiz Time!

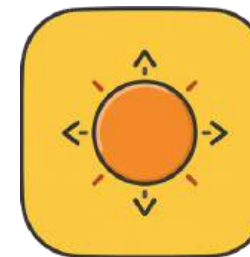
Test Your Knowledge!



Q1: What flows like water through pipes?
A. Electricity



Q2: What does a voltage divider do?
A. Splits electricity to change brightness



Q3: How do you make the light brighter?
A. Turn the potentiometer knob more!



Electricity Flows Like Water Through Pipes!



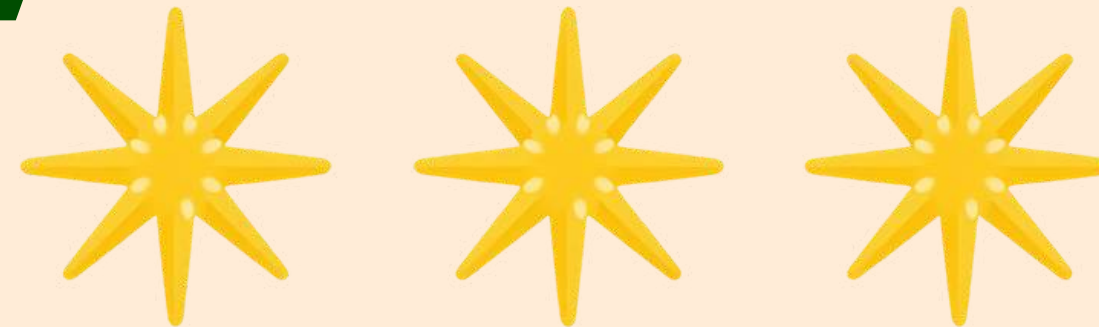
Voltage Divider: Splits electricity to make things brighter or dimmer!



Potentiometer: A special knob you can turn to control the light!

Recap - What We Learned Today

Ready to try your own manual dimmer?



THANK YOU And Make Sure Safety First!



Always ask an adult for help before starting any experiment.



Never touch wires when they are plugged into a power source!



Use batteries, not wall plugs, for all your electricity experiments.